Datasets Report

1. ADL Activities – No Errors

> 20 participants performing 5 Activities of Daily Living (ADL).

Files follow the naming convention: `participant.task.csv`.

Tasks

1. Make a phone call.
2. Wash hands.
3. Cook.
4. Eat.
5. Clean.

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2. ADL Activities – With Errors

> This dataset follows the same structure and activities as the **ADL Activities – No Errors** dataset, but introduces specific mistakes during task execution.

Tasks and Associated Errors

1. **Make a phone call:** The participant initially dials the wrong number and has to redial.
2. **Wash hands:** The participant does not turn off the water after washing their hands.
3. **Cook:** The participant does not turn off the burner after cooking the oatmeal.
4. **Eat:** The participant forgets to bring the medicine container to the dining room.
5. **Clean:** The participant does not use water when cleaning the dishes.

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3. Interweaved ADL Activities

> This dataset follows the same general structure as the previous datasets but introduces additional activities that are interwoven with the primary tasks. Additional files are provided to represent these interwoven activities for each participant.

Tasks

1. Fill medication dispenser.
2. Watch DVD.
3. Water plants.
4. Answer the phone.
5. Prepare birthday card.
6. Prepare soup.
7. Clean.
8. Choose outfit.

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4. Multi-Resident ADL Activities

> Contains data from 26 pairs of participants. Each file represents the activities of a pair of residents.

File Structure

Each record follows this structure:

`[Date][Time][Sensor ID][State][Resident ID][Activity ID]`

When both residents perform activities simultaneously, the last two columns are repeated to represent both residents and their respective activities.

Tasks

1. **Fill medication dispenser (Person A):** Fill the medication dispenser in the kitchen using items obtained from the cabinet. Return all items to the cabinet when finished.
2. **Hang up clothes (Person B):** Hang up clothes in the hallway closet. The clothes are initially placed on the living room couch.

3. **Move furniture (Person B):** Move the couch and coffee table to the other side of the living room. Request assistance from Person A, who temporarily stops filling the medication dispenser to help and resumes the task afterward.
4. **Read a magazine (Person B):** Sit on the couch and read a magazine.
5. **Water plants (Person A):** Water the plants located around the apartment using the watering can from the hallway closet. Return the watering can when finished.
6. **Sweep the kitchen (Person B):** Sweep the kitchen floor using the broom and dustpan located in the kitchen closet. Return the tools when finished.
7. **Play checkers (Persons A and B):** Play a game of checkers for a maximum of five minutes.
8. **Prepare dinner ingredients (Person A):** Set out the ingredients for dinner in the kitchen.
9. **Set the dining table (Person B):** Prepare the dining room table for dinner.
10. **Read a magazine (Person A):** Read a magazine while sitting on the living room couch.
11. **Simulate paying an electric bill (Person B):** Retrieve a check, pen, and envelope from the cupboard underneath the living room television. Use the telephone book in the dining room to look up a utility company's phone number and confirm the bill amount. Request assistance from Person A if needed. Person A temporarily stops their current task to help and resumes it afterward.
12. **Gather picnic food (Person A):** Collect food from the kitchen cupboard and pack it into a picnic basket.
13. **Retrieve dishes (Person B):** Retrieve dishes from a kitchen cabinet. Request assistance from Person A to identify the correct cabinet if needed. Person A temporarily stops their current task to help and resumes it afterward.
14. **Pack picnic supplies (Person B):** Pack the necessary supplies into the picnic basket.
15. **Pack food and move the basket (Person A):** Pack the food into the picnic basket and bring it to the apartment's front door.

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5. Main Daily Life Activities

> This dataset contains sensor data collected from 189 homes over an 18-year period (2007–2024). Labeled activity data is available for 82 homes.

Activities in the Labeled Data

1. Step_Out
2. Toilet
3. Phone
4. Personal_Hygiene
5. Leave_Home
6. Enter_Home
7. Relax
8. Sleep_Out_Of_Bed
9. Drink
10. Watch_TV
11. Dress
12. Evening_Meds
13. Wake_Up
14. Read
15. Morning_Meds
16. Cook_Breakfast
17. Eat_Breakfast
18. Bathe
19. Cook_Lunch
20. Eat_Lunch
21. Wash_Lunch_Dishes
22. Go_To_Sleep
23. Sleep

24. Bed_Toilet_Transition
25. Wash_Breakfast_Dishes
26. Work_At_Table
27. Groom
28. Cook
29. Eat
30. Cook_Dinner
31. Eat_Dinner
32. Wash_Dinner_Dishes
33. Wash_Dishes
34. Entertain_Guests

Common Dataset Information

Sensors

The datasets include several types of sensors deployed throughout the apartment.

Sensor ID	Sensor Type	Description
M01–M51	PIR Motion Detectors	Detect movement, reporting ON when motion starts and OFF when it stops.
I01–I08	Item Use Sensors	Detect the presence or absence of specific objects on the sensors.
D01–D12	Door Sensors	Monitor cabinet doors, reporting whether they are open or closed.
AD1-A / AD1-B	Water Sensors	Monitor water levels in the kitchen sink.
AD1-C	Burner Sensor	Monitor the burner's activity level.
P01	Phone Use Sensor	Detect phone usage.

Sensor ID	Sensor Type	Description
T01–T03	Temperature Sensors	Monitor temperature changes.

Item Use Sensors (I01–I08)

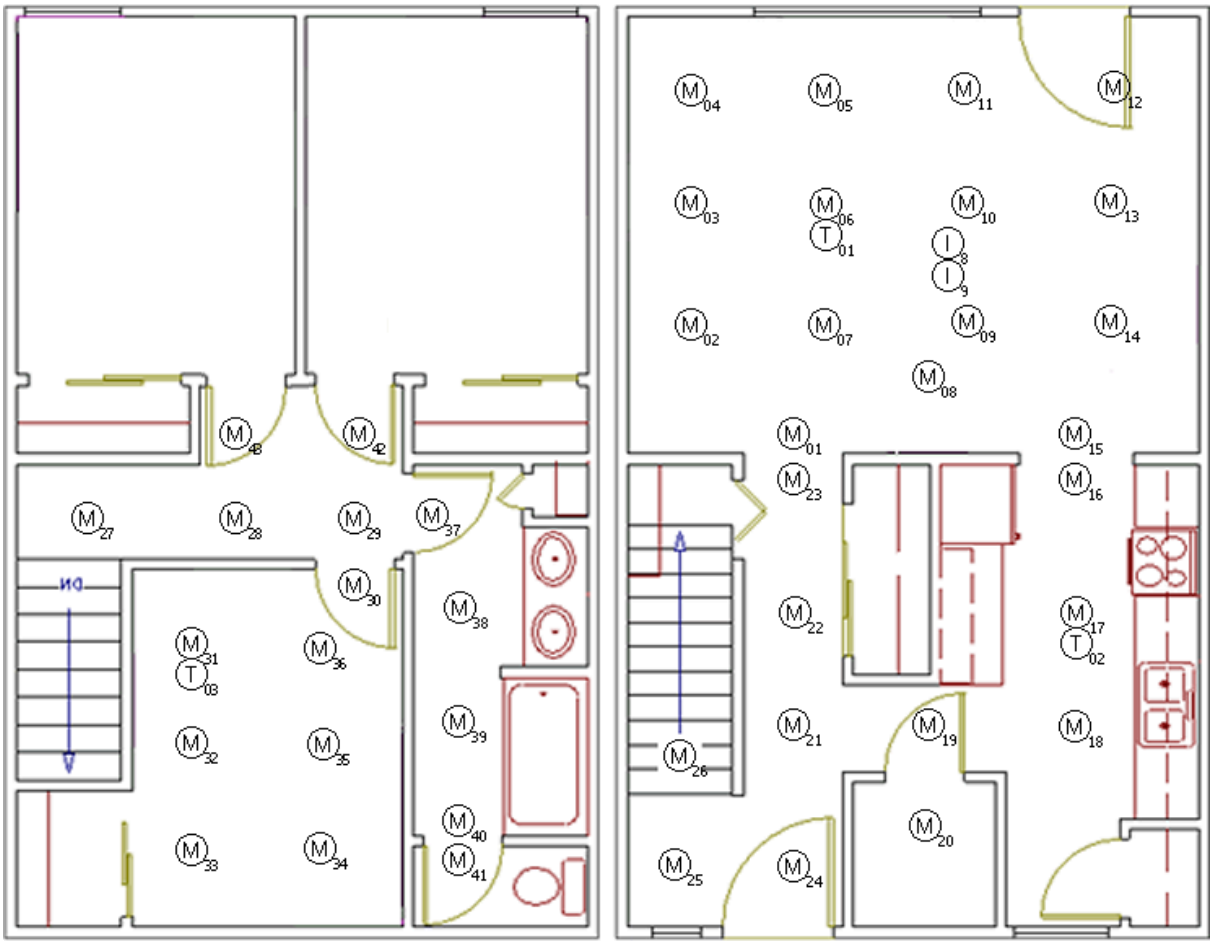
The item use sensors correspond to the following objects, respectively:

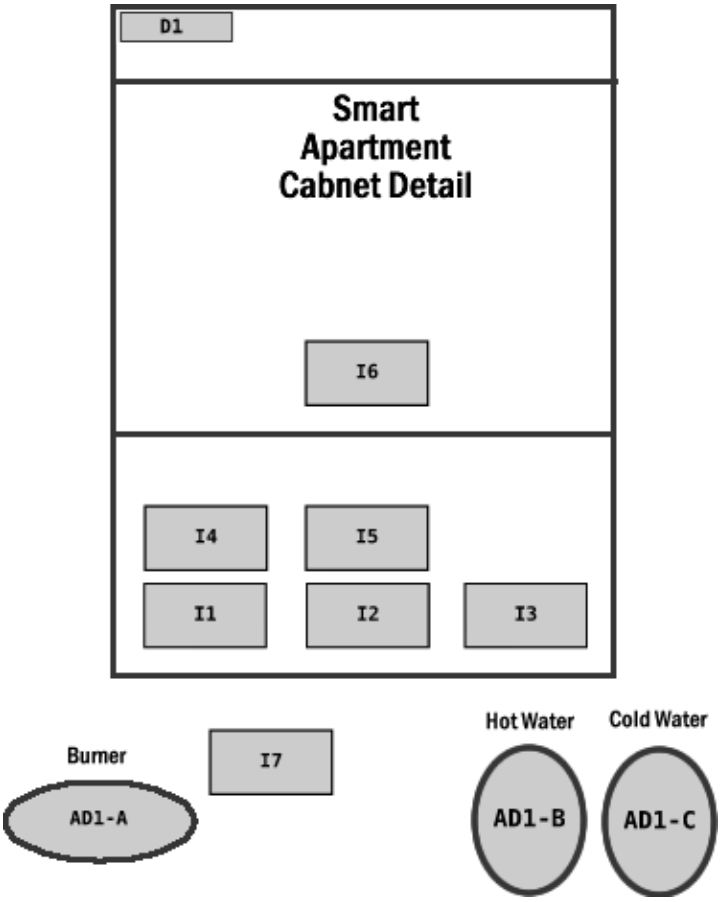
1. Oatmeal
2. Raisins
3. Brown sugar
4. Bowl
5. Measuring spoon
6. Medicine container
7. Pot
8. Phone book

Note: These sensors indicate whether an item is present on or absent from its corresponding sensor.

Floor Plans

The datasets also include floor plan images illustrating the apartment layout and the locations where the sensors are deployed.


M04
M05
M11
M12
M03
M06
T01
M10
M13
I8
I9
M02
M07
M09
M14
M08
M48
M42
M01
M15
M23
M16
M27
M28
M29
M37
M30
M38
M22
M17
T02
M31
T08
M36
M39
M21
M19
M18
M32
M35
M26
M40
M20
M33
M34
M41
M25
M24


D1
Smart
Apartment
Cabnet Detail
I6
I4
I5
I1
I2
I3
Burner
I7
AD1-A
Hot Water
Cold Water
AD1-B
AD1-C